

JAVA OOP PRACTICAL RECORD

Student Utility Program

Experiment No.	01
Date	29/08/2026
Title	Implementation of a Menu-Driven Student Utility Program in Java

1. Objective / Aim

To write a Java menu-driven program that performs common student utility operations such as student information processing, even-or-odd checking, largest-number calculation, grade determination, day selection, multiplication table generation, number series, factorial, prime checking, maximum of two numbers, and area of a circle.

2. System Requirements

Hardware: Standard PC/Laptop with Core i3/i5 processor or equivalent and 4 GB/8 GB RAM.

Software: Windows/Linux/macOS, JDK 17 or JDK 21, and a text editor or IDE such as VS Code, Eclipse, IntelliJ IDEA, or NetBeans.

3. Algorithm / Logic Description

1. Import Scanner for keyboard input.
2. Create the StudentUtilityProgram class and utility methods for factorial, prime checking, maximum, and circle area.
3. Create a Scanner object and display a menu inside a do-while loop.
4. Read the user's choice.
5. Use switch-case to execute the selected operation.
6. Read required values and display the calculated result.
7. Repeat until the user selects option 0.
8. Close the Scanner and terminate the program.

4. UML Class Diagram

StudentUtilityProgram

Methods:

```
+ factorial(n : int) : int
+ prime(n : int) : boolean
+ max(a : int, b : int) : int
+ areaOfCircle(r : double) : double
+ main(args : String[]) : void
```

Uses: Scanner

Relationship: main() creates an object of StudentUtilityProgram and calls the utility methods.

5. Source Code

```
package task1;
```

```

import java.util.Scanner;

public class StudentUtilityProgram {

    int factorial(int n) {
        int fact = 1;
        for (int i = 1; i <= n; i++) {
            fact *= i;
        }
        return fact;
    }

    boolean prime(int n) {
        if (n <= 1) return false;

        for (int i = 2; i < n; i++) {
            if (n % i == 0) return false;
        }
        return true;
    }

    int max(int a, int b) {
        return (a > b) ? a : b;
    }

    double areaOfCircle(double r) {
        return Math.PI * r * r;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int choice;
        StudentUtilityProgram util = new StudentUtilityProgram();

        do {
            System.out.println("\n===== Student Utility Program =====");
            System.out.println("1. Student Information");
            System.out.println("2. Even or Odd");
            System.out.println("3. Largest of Three Numbers");
            System.out.println("4. Grade based on percentage");
            System.out.println("5. Day of a week");
            System.out.println("6. Multiplication Table");
            System.out.println("7. Numbers from 1 to N");
            System.out.println("8. Sum of first N natural numbers");
            System.out.println("9. Fibonacci Series");
            System.out.println("10. Factorial of a number");
            System.out.println("11. Prime number check");
            System.out.println("12. Max of Two Numbers");
            System.out.println("13. Area of Circle");
            System.out.println("0. Exit");

            System.out.print("Enter your choice: ");
            choice = sc.nextInt();

            switch (choice) {
                case 1:
                    sc.nextLine();
                    System.out.print("Enter student name: ");
                    String name = sc.nextLine();

                    System.out.print("Enter student roll number: ");
                    int rollNumber = sc.nextInt();

                    System.out.print("Enter student marks in 3 subjects: ");
                    int m1 = sc.nextInt();
                    int m2 = sc.nextInt();
                    int m3 = sc.nextInt();

                    int totalMarks = m1 + m2 + m3;
                    double percentage = totalMarks / 3.0;

                    System.out.println("\nStudent Information:");
                    System.out.println("Name: " + name);
                    System.out.println("Roll Number: " + rollNumber);
                    System.out.println("Total Marks: " + totalMarks);
                    System.out.println("Percentage: " + percentage);
                    break;

                case 2:
                    System.out.print("Enter a number: ");
                    int num = sc.nextInt();
                    System.out.println(num % 2 == 0
                        ? num + " is even."
                        : num + " is odd.");
                    break;

                case 3:
                    System.out.print("Enter three numbers: ");
                    int a = sc.nextInt();
                    int b = sc.nextInt();
                    int c = sc.nextInt();
            }
        } while (choice != 0);
    }
}

```

```

        if (a >= b && a >= c)
            System.out.println(a + " is the largest number.");
        else if (b >= c)
            System.out.println(b + " is the largest number.");
        else
            System.out.println(c + " is the largest number.");
        break;

case 4:
    System.out.print("Enter percentage: ");
    double per = sc.nextDouble();

    if (per >= 90) System.out.println("Grade: A");
    else if (per >= 80) System.out.println("Grade: B");
    else if (per >= 70) System.out.println("Grade: C");
    else if (per >= 60) System.out.println("Grade: D");
    else System.out.println("Grade: F");
    break;

case 5:
    System.out.print("Enter day number (1-7): ");
    int day = sc.nextInt();

    switch (day) {
        case 1: System.out.println("Monday"); break;
        case 2: System.out.println("Tuesday"); break;
        case 3: System.out.println("Wednesday"); break;
        case 4: System.out.println("Thursday"); break;
        case 5: System.out.println("Friday"); break;
        case 6: System.out.println("Saturday"); break;
        case 7: System.out.println("Sunday"); break;
        default: System.out.println("Invalid day number.");
    }
    break;

case 6:
    System.out.print("Enter a number for multiplication table: ");
    int n = sc.nextInt();
    for (int i = 1; i <= 10; i++)
        System.out.println(n + " x " + i + " = " + (n * i));
    break;

case 7:
    System.out.print("Enter a number N: ");
    int N = sc.nextInt();
    for (int i = 1; i <= N; i++)
        System.out.print(i + " ");
    System.out.println();
    break;

case 8:
    System.out.print("Enter a number N: ");
    int N1 = sc.nextInt();
    int sum = 0;
    for (int i = 1; i <= N1; i++) sum += i;
    System.out.println("Sum: " + sum);
    break;

case 9:
    System.out.print("Enter a number N: ");
    int N2 = sc.nextInt();
    int x = 0, y = 1;
    while (x <= N2) {
        System.out.print(x + " ");
        int next = x + y;
        x = y;
        y = next;
    }
    System.out.println();
    break;

case 10:
    System.out.print("Enter a number: ");
    int N3 = sc.nextInt();
    System.out.println("Factorial: " + util.factorial(N3));
    break;

case 11:
    System.out.print("Enter a number: ");
    int N4 = sc.nextInt();
    System.out.println(util.prime(N4)
        ? N4 + " is a prime number."
        : N4 + " is not a prime number.");
    break;

case 12:
    System.out.print("Enter two numbers: ");
    int N5 = sc.nextInt();
    int N6 = sc.nextInt();
    System.out.println("Maximum: " + util.max(N5, N6));
    break;

```

```

        case 13:
            System.out.print("Enter radius of the circle: ");
            double r = sc.nextDouble();
            System.out.println("Area: " + util.areaOfCircle(r));
            break;

        case 0:
            System.out.println("Exiting the program.");
            break;

        default:
            System.out.println("Invalid choice. Please try again.");
    }
} while (choice != 0);

sc.close();
}
}

```

6. Compilation & Execution Commands

Because the program uses package task1, save the file inside a folder named task1.

```
task1/StudentUtilityProgram.java
```

Compile from the parent folder:

```
javac task1/StudentUtilityProgram.java
```

Run the program:

```
java task1.StudentUtilityProgram
```

7. Output Screen

Representative output:

```
===== Student Utility Program =====
```

```
1. Student Information
```

```
2. Even or Odd
```

```
...
```

```
13. Area of Circle
```

```
0. Exit
```

```
Enter your choice: 10
```

```
Enter a number: 5
```

```
Factorial: 120
```

```
Enter your choice: 11
```

```
Enter a number: 17
```

```
17 is a prime number.
```

```
Enter your choice: 0
```

```
Exiting the program.
```

8. Result

The Student Utility Program was successfully implemented using Java. The program demonstrates Scanner input, methods, loops, conditional statements, switch-case, object creation, and menu-driven execution.